



 COMPUTER VISION COURSE

Object Detection Architectures

YOLOv8 · Faster R-CNN · SSD

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Why Object Detection?

Object detection is one of the **core tasks in computer vision**, where AI systems identify objects in images or videos and precisely locate them using **bounding boxes**. It combines both classification and localization in a unified pipeline.

Real-World Applications



Autonomous Vehicles



Healthcare Imaging



Surveillance



Robotics



Sports Analytics



Real-Time AI

Why We Chose This Track

- Directly relates to the project implementation using **YOLOv8 pose estimation**
- Demonstrates the critical **speed vs. accuracy tradeoff** in modern architectures
- Real-time **exercise analysis** depends on fast and efficient detection pipelines
- Highly relevant to **modern AI and computer vision** industry applications

Detection Paradigms: One-Stage vs. Two-Stage

One-Stage Detectors

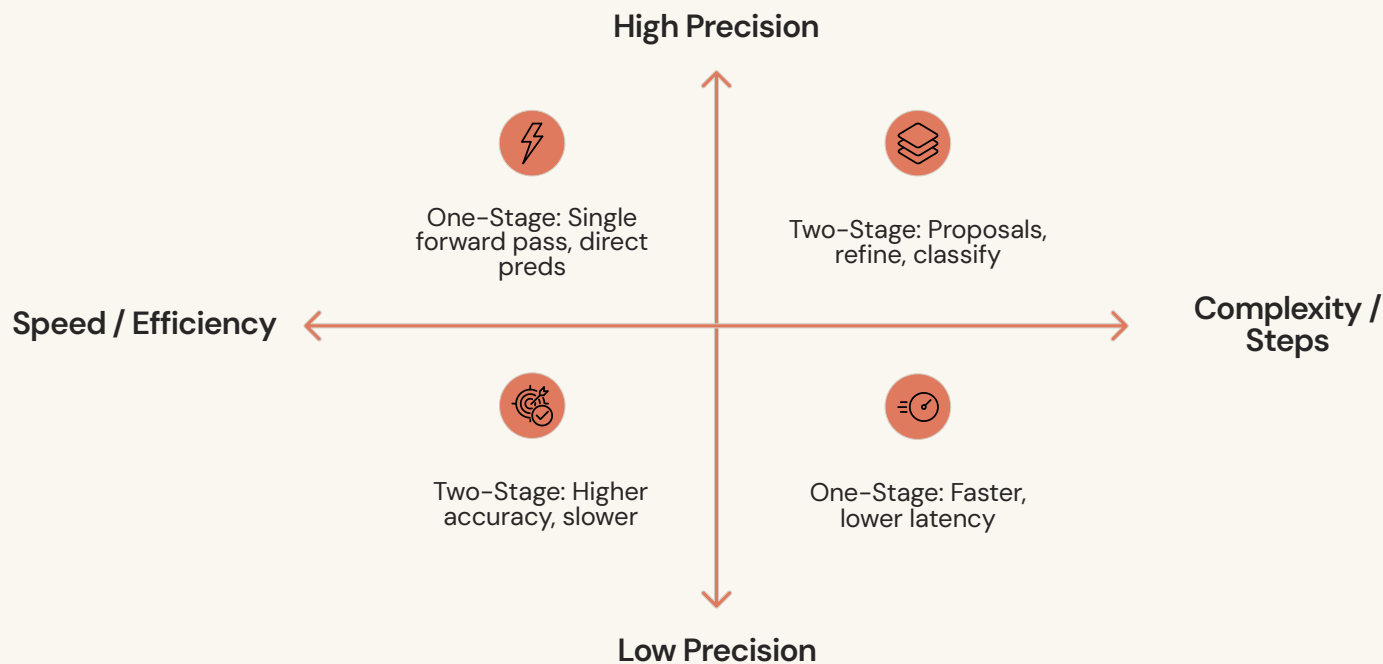
Perform detection in a **single forward pass** through the network. The model simultaneously predicts bounding boxes and class probabilities across the entire image at once.

- Faster inference speed
- Suitable for real-time systems
- Examples: YOLOv8, SSD

Two-Stage Detectors

First generate **region proposals** (candidate object locations), then classify and refine each proposal in a second dedicated stage.

- Higher localization accuracy
- Slower inference speed
- Example: **Faster R-CNN**



Understanding this distinction is fundamental — it explains the core **speed-accuracy tradeoff** that governs which architecture is appropriate for a given task.

YOLOv8 (You Only Look Once)

1

Backbone (Feature Extraction)

A deep CNN (e.g., CSPDarknet) extracts hierarchical feature representations from the input image at multiple scales.

2

Neck (Multi-Scale Feature Fusion)

A PANet/FPN-style neck aggregates features from different backbone layers, enabling detection of objects at varying sizes.

3

Detection Head (Direct Predictions)

A decoupled head simultaneously predicts **bounding boxes coordinates**, **object confidence**, and **class probabilities** in a single forward pass – no region proposals needed.

✓ Advantages

- Extremely fast inference
- Strong speed—accuracy balance
- Efficient for video and live systems

⚠ Limitations

- May slightly sacrifice localization precision vs. slower models
- May miss tiny objects

🔬 Applications

- Surveillance · Autonomous driving
- Pose estimation · Sports tracking · Robotics

Faster R-CNN (Region-Based Detection)

Faster R-CNN introduces the **Region Proposal Network (RPN)**, which shares convolutional features with the detection head, making it faster than its predecessors while remaining accurate.

⚙️ Stage 1 – Proposals

- RPN scans feature map
- Generates candidate object regions
- Assigns objectness scores

⚙️ Stage 2 – Classification

- ROI Pooling extracts fixed-size features
- Fully connected layers classify each region
- Bounding boxes are refined

✅ Advantages

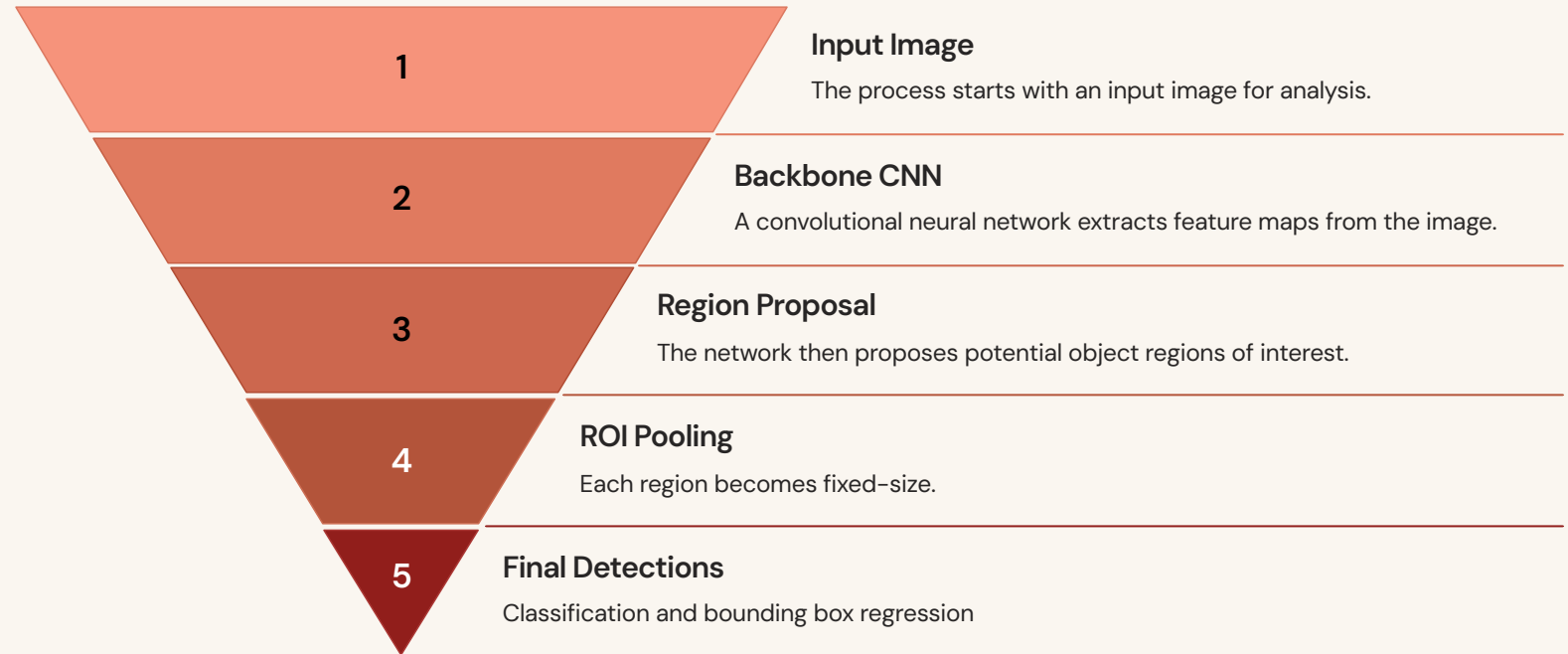
- Highly accurate
- Strong localization quality
- Effective in complex scenes

⚠️ Limitations

- Slow inference speed
- Computationally expensive (Two-stage pipeline)
- Not suited for real-time use

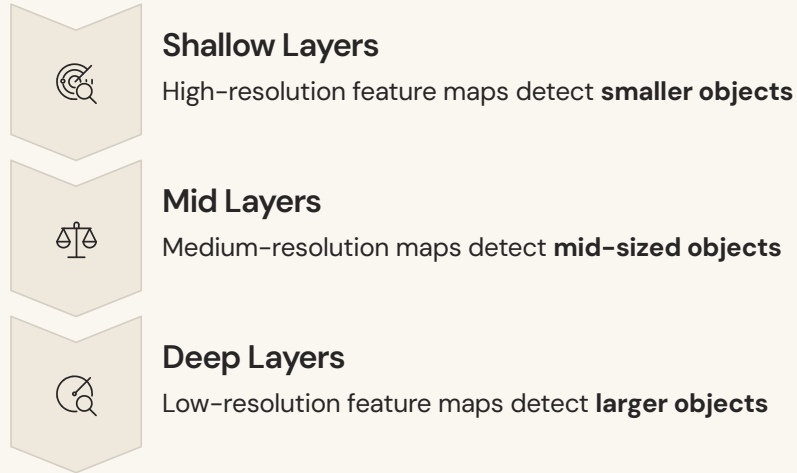
🔬 Applications

- Medical imaging · Industrial inspection
- Research · High-precision detection



SSD (Single Shot Detector)

SSD performs object detection across **multiple convolutional feature maps** of varying resolutions. Different CNN layers are responsible for detecting objects of different sizes.



Anchor Boxes

At each feature map location, SSD places predefined anchor boxes with different aspect ratios and scales to predict objects efficiently. These anchors are crucial to SSD's multi-scale detection approach.

✓ Advantages

- Fast inference speed
- Lightweight architecture
- Suitable for mobile and embedded devices

⚠ Limitations

- Weaker performance on very small objects
- Older architecture vs. modern YOLO variants

🔬 Applications

- Mobile AI · Embedded vision devices
- Lightweight real-time detection systems

Architecture Comparison

Criterion	YOLOv8	Faster R-CNN	SSD
Detector Type	One-Stage	Two-Stage	One-Stage
Structure	Backbone + Neck + Decoupled Head	Backbone + RPN + ROI Pooling + Head	Base CNN + Multi-Scale Feature Maps
Prediction	Direct single forward pass	Region proposals then classification	Simultaneous across multi-scale maps
Speed	⚡ Very Fast	🐢 Slow	⚡ Fast
Accuracy	🎯 High	🎯 Very High	♦ Moderate
Complexity	Medium	High	Low-Medium
Real-Time	✅ Excellent	❌ Not suited	✅ Good
Key Strength	Best speed & accuracy balance	Highest localization quality	Lightweight, mobile-friendly
Key Weakness	Slight localization tradeoff	Slow, computationally expensive	Weak on very small objects
Best For	Real-time video, pose estimation	Medical imaging, precision tasks	Mobile AI, embedded systems

YOLOv8

Best balance of speed & accuracy for real-time systems

Faster R-CNN

Prioritizes accuracy over inference speed

SSD

Lightweight and efficient, but lower performance vs. modern YOLO

✓ CONCLUSION

Why YOLOv8 for the Exercise Analysis Project?



Fast Inference

YOLOv8 processes frames in real-time, making it ideal for **live video analysis** without perceptible latency.



Strong Detection

Delivers a **high-accuracy detection pipeline** capable of reliably identifying body keypoints and poses across frames.



Pose Estimation Ready

YOLOv8 natively supports **pose estimation tasks**, providing an end-to-end solution for exercise tracking and movement analysis.



Efficient Pipeline

Its single-stage architecture ensures **low computational overhead**, suitable for deployment on standard hardware and embedded systems.

Among the three architectures studied, **YOLOv8 is the optimal choice** for this project → delivering the fast inference, strong detection performance, and efficient real-time processing essential for pose estimation and video-based exercise analysis.